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# REINFORCEMENT LEARNING

REINFORCEMENT LEARNING
TOPIC OUTLINE


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UNIT 3

# BANDIT PROBLEMS

# Recommended sources:

- https://www.davidsilver.uk/teaching/

  on-line lectures (videos + slides)

- https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf

  on-line book

# Exploration and exploitation

## EXPLORATION AND EXPLOITATION

Exploration

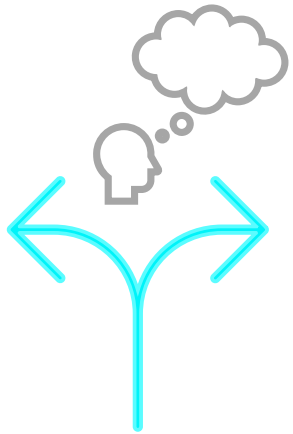
— find as much as possible information about the environment

Exploitation

— exploit known information to maximise the reward

both are important

EXPLORATION VS EXPLOITATION DILLEMA



## Exploitation

- Choose high reward actions
- Exploit learned information
- Maximize immediate gain

## Exploration

- Discover novel actions
- Learn about environment
- Balance knowledge expansion

When to explore new options and when to exploit what we know?

**Dilemma:** balancing immediate gains vs. exploration for long-term optimal choices in sequential decision-making

- **Scenario:** conduct clinical trials for multiple treatments



- **Problem:** how to balance optimal individual patient care and discover the overall best medical treatment?





How much to explore / exploit?

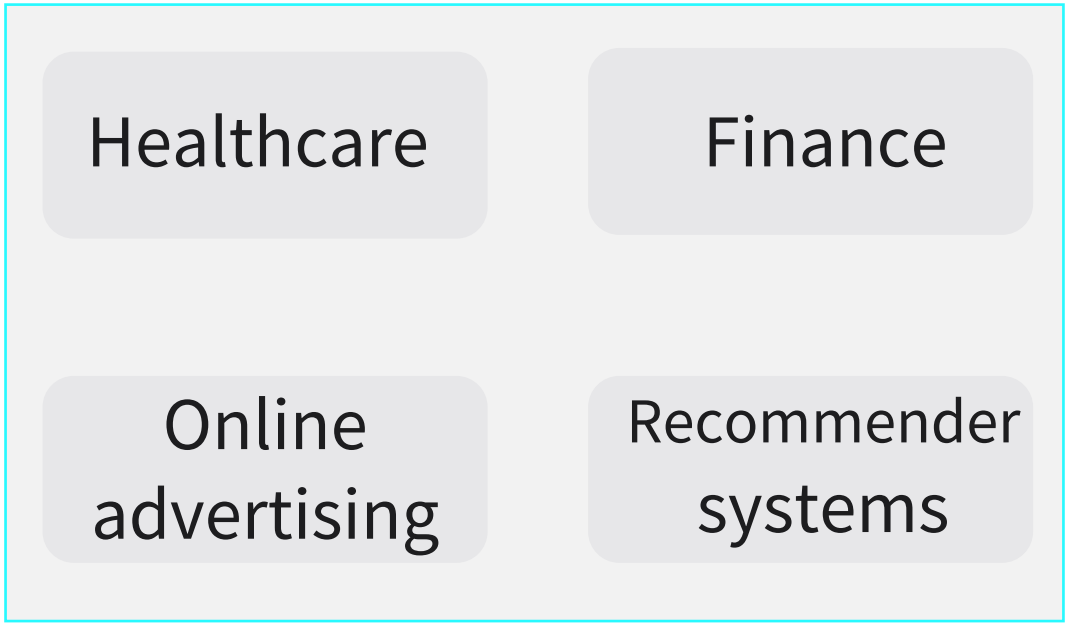
Source of the text: Freedman, 1987



**Goal:** balance optimal decisions with exploration in sequential processes



- **Adaptation**: optimise decisions based on evolving data
- **Iterative Learning**: continuously refine strategies for maximised gains




Applications

# Exploration-exploitation strategies

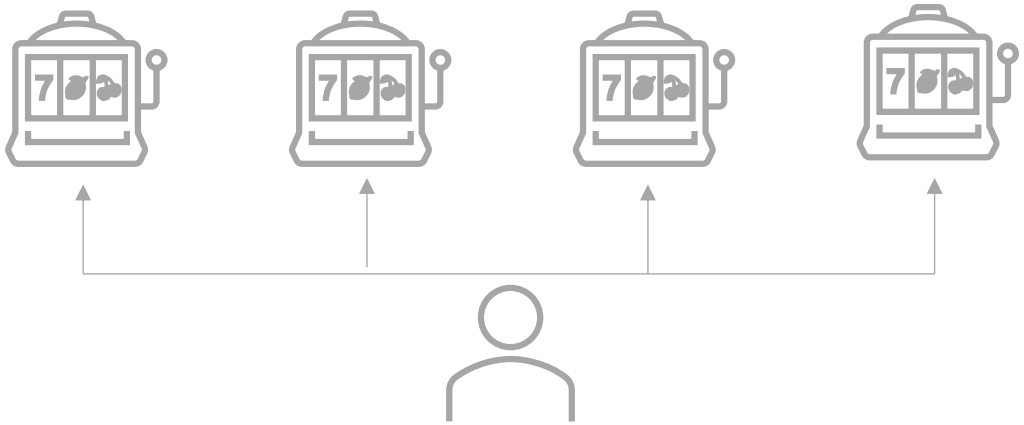
- Random exploration
- “optimism in the face of uncertainty”
  preference to exploring uncertain states/actions
- Information state space

# Bandit problems

# Multiple arm bandits

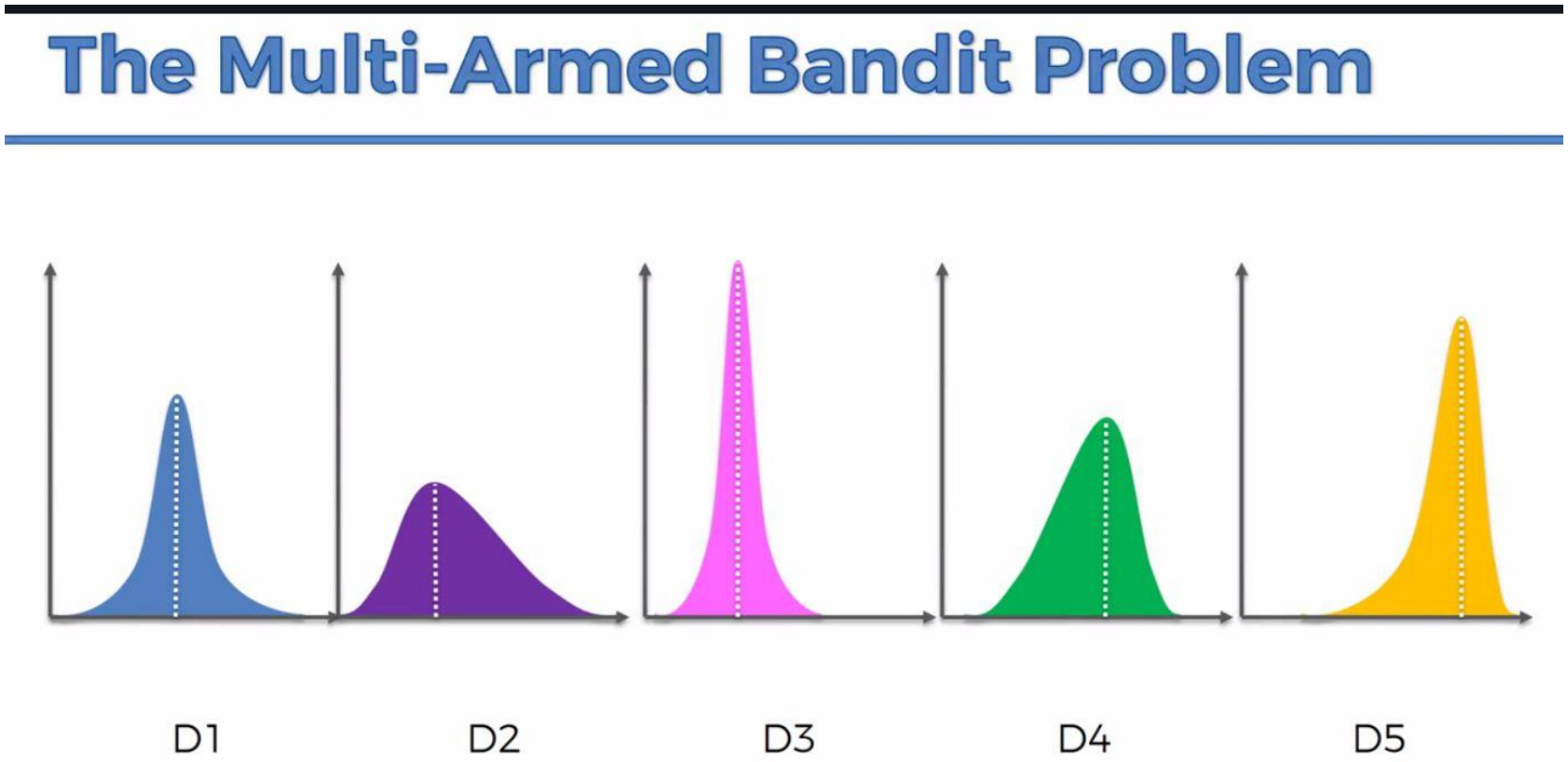
**Multiple Arm Bandit:** optimise choices among multiple actions to maximise cumulative rewards

- **Multiple Arms:** set of actions
- **Reward probability distribution: unknown**
  uncertain rewards for each action
- **Initial Knowledge**: limited or no reward distribution information
- **Learning**: update action preferences based on observed rewards over time



Maximise rewards:
How to choose actions?

probability distributions of the rewards for each action (unknown)



# Multiple arm bandit

- a tuple $(A, R)$
- $A$ = a known set of actions ("arms")
- $R^a(r) = P(r|a)$ unknown probability distribution of rewards for each action $a$
- $a_t$ = action taken at the time step $t$
- $r_t$ = reward obtained at the time step $t$

- The goal: maximise the cumulative reward

$$R = \sum_{\tau=1}^{t} r_\tau$$

# Regret

- Mean reward for action $a$: $q(a) = E[r|a]$
- Optimal value: $v^* = q(a^*) = \max_a q(a)$
- Regret at the time step $t$: $l_t = E[v^* - q(a_t)]$
- Total regret:

$$L_t = E\left[\sum_{\tau=1}^{t}(v^* - q(a_\tau))\right]$$

maximise cumulative reward = minimise total regret

# Regret: a function of counts and differences (gaps)

$$L_t = E\left[\sum_{\tau=1}^{t}(v^* - q(a_\tau))\right] =$$

$$\sum_a E[N_t(a)] \cdot (v^* - q(a)) \; = \; \sum_a E[N_t(a)] \cdot \Delta(a)$$

$N_t(a) =$ number of times action $a$ was selected until the time step $t$

$\Delta(a) =$ difference between values of the optimal action $a^*$ and selected action $a$

$\Delta(a)$ unknown!

# Regret

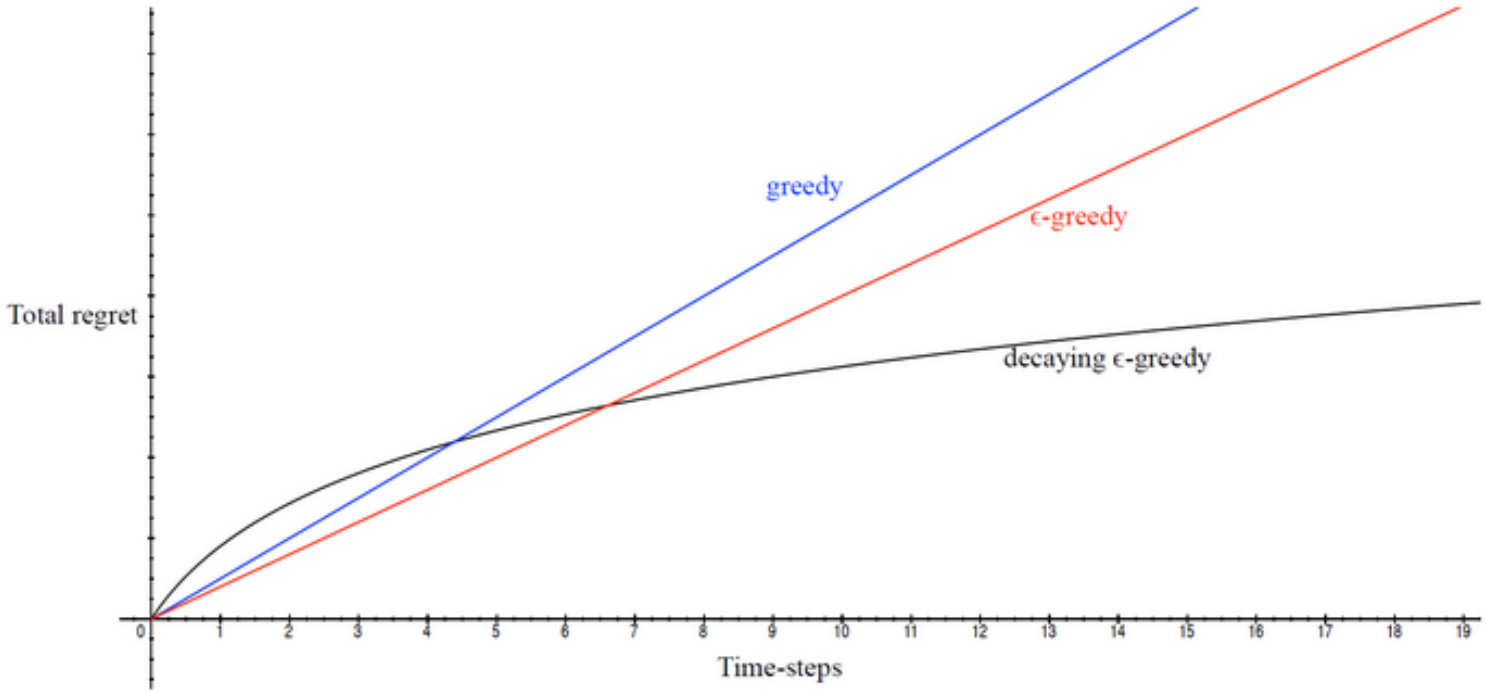
$$L_t = \sum_a E[N_t(a)] \cdot \Delta(a)$$

increases over time

depending on the strategy, increases linearly or sub-linearly

# Regret

depending on the strategy, increases linearly or sub-linearly



# Exploitation-exploration strategies

# Exploitation-exploration strategies for bandit problems

random

- greedy strategy (pure exploitation)
- $\varepsilon$-greedy strategy (combining exploitation and exploration)

optimism and uncertainty

- UCB -- upper confidence bound (better combining exploitation and exploration)

information state space

- introducing states as summaries of all accumulated information

# Greedy strategy

- estimate the value of each action $\hat{q}_t(a) = \frac{\#\text{times action } a \text{ is selected and won reward}}{\#\text{times action } a \text{ is selected}}$
- select the action with the highest value

$$a_t^* = \underset{a}{\operatorname{argmax}}\, \hat{q}_t(a)$$

can get stuck on a sub-optimal action

linear total regret

# Greedy strategy: getting stuck

- A giving a reward with $50\%$ chance
- B giving a reward with $80\%$ chance

Unlucky sequence of initial events:

1) tried A and got a reward
2) tried B and did not get a reward

$\Rightarrow$ keep pulling A

# $\varepsilon$-greedy strategy

- select a random action with probability $\varepsilon$
- select the action with the highest value (as in greedy strategy) with probability $1 - \varepsilon$

$\varepsilon = 1 \quad \Rightarrow \quad$ pure exploration

$\varepsilon = 0 \quad \Rightarrow \quad$ pure exploitation (greedy strategy)

again a linear total regret

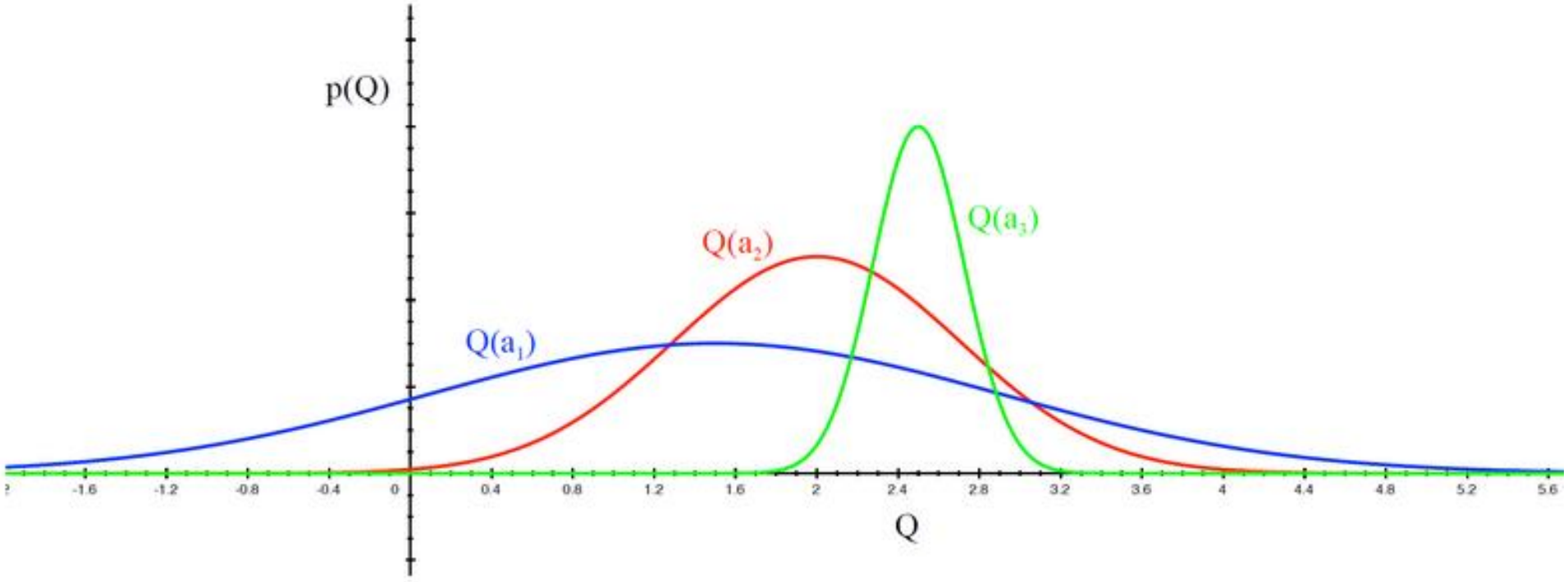
# $\varepsilon_t$-greedy strategy

$\varepsilon$ not constant but decreasing with time

- logarithmic total regret 🙂
- requires knowledge about rewards/gaps 😕

# Upper Confidence Bound (UCB)

three possible actions with probability distributions (unknown to the agent)



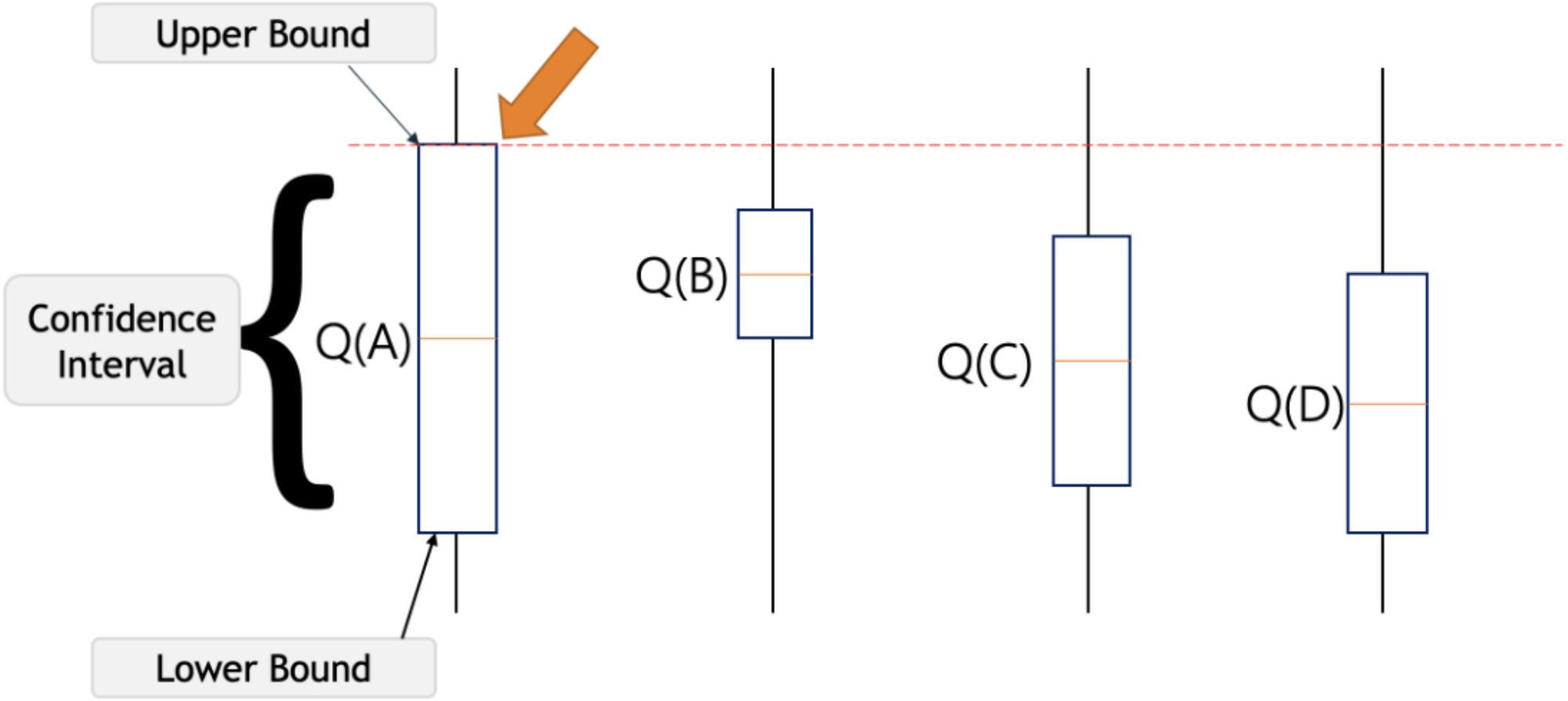
# Upper Confidence Bound (UCB)

- estimate upper confidence for each action value $\hat{u}_t(a)$
  so that $q(a) \leq \hat{q}_t(a) + \hat{u}_t(a)$ with a high probability

- select the action which maximises upper confidence bound

$$a_t^* = \underset{a}{\operatorname{argmax}}\left(\hat{q}_t(a) + \hat{u}_t(a)\right)$$

# Upper Confidence Bound (UCB)



# Upper Confidence Bound (UCB)

upper confidence depends on how many times the action has been selected

- small $N_t(a) \Rightarrow$ high $\hat{u}_t(a) \Rightarrow$ *high uncertainty*
- large $N_t(a) \Rightarrow$ low $\hat{u}_t(a) \Rightarrow$ *low uncertainty*

exploring the action $a$ more and more decreases the value of $\hat{u}_t(a)$

$\Rightarrow$ agent becomes more certain about the choices

# Upper Confidence Bound (UCB)

$$a_t^* = \underset{a}{\operatorname{argmax}} \left( \hat{q}_t(a) + \sqrt{\frac{2 \ln t}{N_t(a)}} \right)$$

no knowledge about the rewards (differences, gaps) needed

# Upper Confidence Bound (UCB)

Hoeffding's inequality

$$P[E[X] > \bar{X}_t + u] \leq e^{-2tu^2}$$

$$P[q(a) > \hat{q}_t(a) + \hat{u}_t(a)] \leq e^{-2N_t(a){u_t}^2(a)}$$

$p \leq e^{-2N_t(a){u_t}^2(a)}$ probability $p$

$p = t^{-4}$ it should be decreasing over time

# Information state space

- Bandit as a sequential decision problem (not only one-state)
- With an “information state” $\tilde{s}$ at each step
  summarising all information accumulated until this step

Each action causes transition to a new state $\tilde{s}'$ by adding information
with probability $P^{a}_{\tilde{s}\tilde{s}'}$,

$\Rightarrow$ a Markov Decision Process $(\tilde{S}, A, \tilde{P}, R, \beta)$

# Contextual bandits

# Contextual bandit

- a tuple $(A, S, R)$
- $A$ = a known set of actions (“arms”)
- $S = P[s]$ = an unknown distribution of states (contexts)
- $R_s^a(r) = P(r|s,a)$ unknown probability distribution of rewards for each action $a$ and context $s$
- $a_t$ = action taken at the time step $t$
- $s_t$ = state/context at the time step $t$
- $r_t$ = reward obtained at the time step $t$

The goal: maximise the cumulative reward $R = \sum_{\tau=1}^{t} r_\tau$

# Questions?

# Examples

# Tetris

States – screen pixels

Actions: rotate, left, right, drop

Rewards: number of cleared lines

# Recommendation systems

# Bandits

Actions (arms): articles

Rewards: clicks, time

Contextual bandits:

Context: information about the user/article

# Clinical trials

# Bandits

Actions (arms): medicines

Rewards: healed patients

SESSION 3

# TRANSFER TASK



## Case study



An online news platform seeks to maximize user engagement by recommending the most relevant articles while balancing click-through rates and diversity of content.
The task involves selecting articles to display to users in order to optimize engagement and satisfaction, considering variables like user preferences, article popularity, and novelty

## Task

Research how the multi-arm bandit problem can enhance article recommendation. Define arms, rewards, exploration, exploitation

**TRANSFER TASK**
**PRESENTATION OF THE RESULTS**

Please present your results.

The results will be discussed in plenary.





1. five-stage finite horizon four-arm bandit problem uses the following sequence of discount factors: $\rho[k] = 1, 1, 1, 0.5, 0.25$, $k = 1,2,3,4,5$. Suppose the true mean rewards are 24.5, 36.5, 19.0, 32.5, for arms $j = 1,2,3,4$ respectively. What is the maximum possible total reward?
   a) 71.25
   b) 95.0
   c) 136.875
   d) 182.5



2. A three-arm bandit problem is a 100-stage finite horizon process. The true mean rewards of the arms are 10.1, 9.0 and 7.5. Suppose a policy selects each arm for ten stages, calculates the average reward for each, then selects the arm with the highest observed average. If the optimal reward is correctly identified, what will be the regret?
   a) 37.0
   b) 266.0
   c) 973.0
   d) 1010.0



3. For softmax action selection, what best describes the policy as the temperature parameter approaches zero?
   a) An exploratory policy
   b) A greedy policy
   c) An $\epsilon$-greedy policy
   d) A UCB1 policy

## LIST OF SOURCES

Freedman, B. (1987). Equipoise and the ethics of clinical research. New England Journal of Medicine, 317 (3), 141–145.

Sutton, R. S., & Barto, A. G. (2018a). Reinforcement learning: An introduction (2nd ed.). MIT press.

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